



ER Diagram Design and Relational Schema

Online Food Delivery System

Prepared By: Kunguma Keerthana M

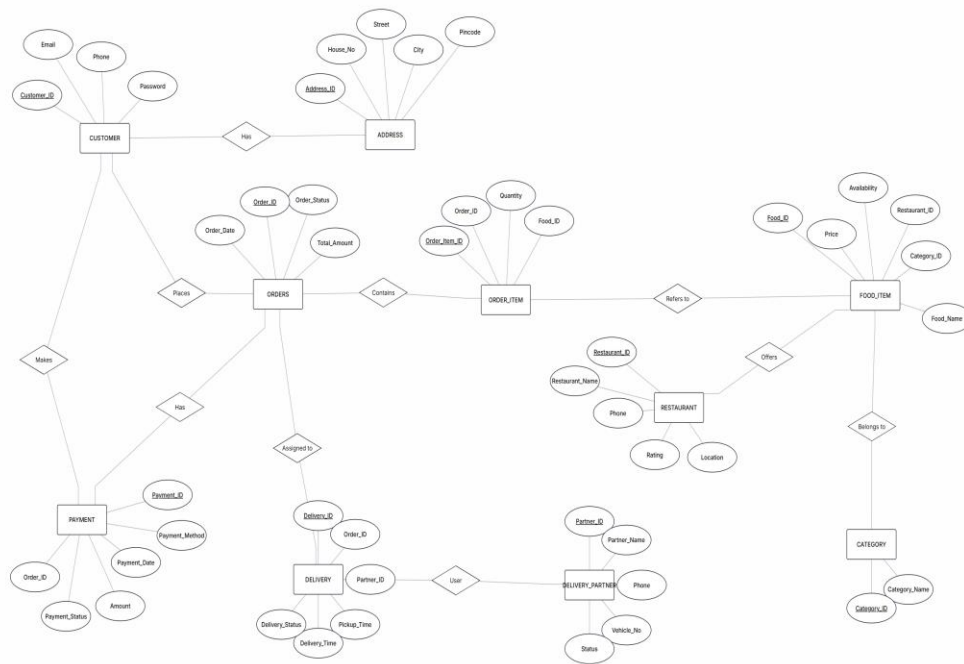
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Subject: Database Management Systems

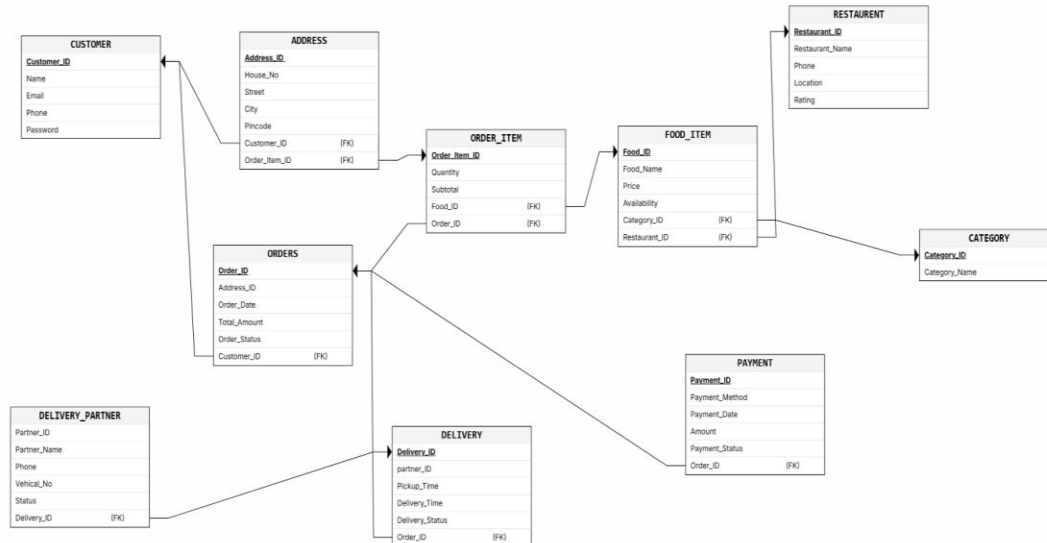
OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Online Food Delivery System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

ER DIAGRAM



RELATIONAL SCHEMA



ENTITIES AND ATTRIBUTES

CUSTOMER

Attribute	Key
Customer_ID	PK
Name	-
Email	-
Phone	-
Password	-

ADDRESS

Attribute	Key
Address_ID	PK
House_No	-
Street	-
City	-
Pincode	-
Customer_ID	FK
Order_Item_ID	FK

ORDERS

Attribute	Key
Order_ID	PK
Address_ID	-
Order_Date	-
Total_Amount	-
Order_Status	-
Customer_ID	FK

ORDER_ITEM

Attribute	Key
Order_Item_ID	PK
Quantity	-
Subtotal	-
Food_ID	FK
Order_ID	FK

FOOD_ITEM

Attribute	Key
Food_ID	PK
Food_Name	-
Price	-
Availability	-
Category_ID	FK
Restaurant_ID	FK

RESTAURANT

Attribute	Key
Restaurant_ID	PK
Restaurant_Name	-
Phone	-
Location	-
Rating	-

CATEGORY

Attribute	Key
Category_ID	PK
Category_Name	-

PAYMENT

Attribute	Key
Payment_ID	PK
Payment_Method	-
Payment_Date	-
Amount	-
Payment_Status	-
Order_ID	FK

DELIVERY_PARTNER

Attribute	Key
Partner_ID	-
Partner_Name	-
Phone	-
Vehicle_No	-
Status	-
Delivery_ID	FK

DELIVERY

Attribute	Key
Delivery_ID	PK
partner_ID	-
Pickup_Time	-
Delivery_Time	-
Delivery_Status	-
Order_ID	FK

RELATIONSHIPS AND CARDINALITIES

Entity 1	Entity 2	Cardinality / Connection
CUSTOMER	ADDRESS	1:M
CUSTOMER	ORDERS	1:M
ADDRESS	ORDER_ITEM	relationship shown by connector
ORDERS	ORDER_ITEM	1:M
FOOD_ITEM	ORDER_ITEM	1:M
RESTAURANT	FOOD_ITEM	1:M
CATEGORY	FOOD_ITEM	1:M
ORDERS	PAYMENT	1:1
ORDERS	DELIVERY	1:1
DELIVERY_PARTNER	DELIVERY	1:M

RELATIONAL SCHEMA

CUSTOMER(Customer_ID PK, Name, Email, Phone, Password)

ADDRESS(Address_ID PK, House_No, Street, City, Pincode, Customer_ID FK, Order_Item_ID FK)

ORDERS(Order_ID PK, Address_ID, Order_Date, Total_Amount, Order_Status, Customer_ID FK)

ORDER_ITEM(Order_Item_ID PK, Quantity, Subtotal, Food_ID FK, Order_ID FK)

FOOD_ITEM(Food_ID PK, Food_Name, Price, Availability, Category_ID FK, Restaurant_ID FK)

RESTAURANT(Restaurant_ID PK, Restaurant_Name, Phone, Location, Rating)

CATEGORY(Category_ID PK, Category_Name)

PAYMENT(Payment_ID PK, Payment_Method, Payment_Date, Amount, Payment_Status, Order_ID FK)

DELIVERY_PARTNER(Partner_ID, Partner_Name, Phone, Vehicle_No, Status, Delivery_ID FK)

DELIVERY(Delivery_ID PK, partner_ID, Pickup_Time, Delivery_Time, Delivery_Status, Order_ID FK)

PRIMARY KEY AND FOREIGN KEY IDENTIFICATION

Table	Primary Key	Foreign Key(s)
CUSTOMER	Customer_ID	-
ADDRESS	Address_ID	Customer_ID, Order_Item_ID
ORDERS	Order_ID	Customer_ID
ORDER_ITEM	Order_Item_ID	Food_ID, Order_ID
FOOD_ITEM	Food_ID	Category_ID, Restaurant_ID
RESTAURANT	Restaurant_ID	-
CATEGORY	Category_ID	-
PAYMENT	Payment_ID	Order_ID
DELIVERY_PARTNER	-	Delivery_ID
DELIVERY	Delivery_ID	Order_ID

RELATIONSHIP EXPLANATION

Customer is connected to Address and Orders. The diagram shows one customer associated with multiple addresses and orders.

Orders are connected to Order_Item, Payment, and Delivery. Order_Item stores the individual food items belonging to an order.

Food_Item is connected to Restaurant and Category, showing that food items are associated with a restaurant and a category.

Delivery_Partner is connected to Delivery, representing assignment of deliveries to delivery partners.

The uploaded ER diagram is used as the source for the table names, attributes, keys, and visible connections in this document.